

南開大學

信息对抗技术课程实验报告

实验四：扫描工具 Nmap 的使用



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一、实验目的

1. 熟悉 NMap 使用方法，熟悉扫描工具的原理。
2. 使用 NMap 扫描网站。

二、实验原理

1、Nmap

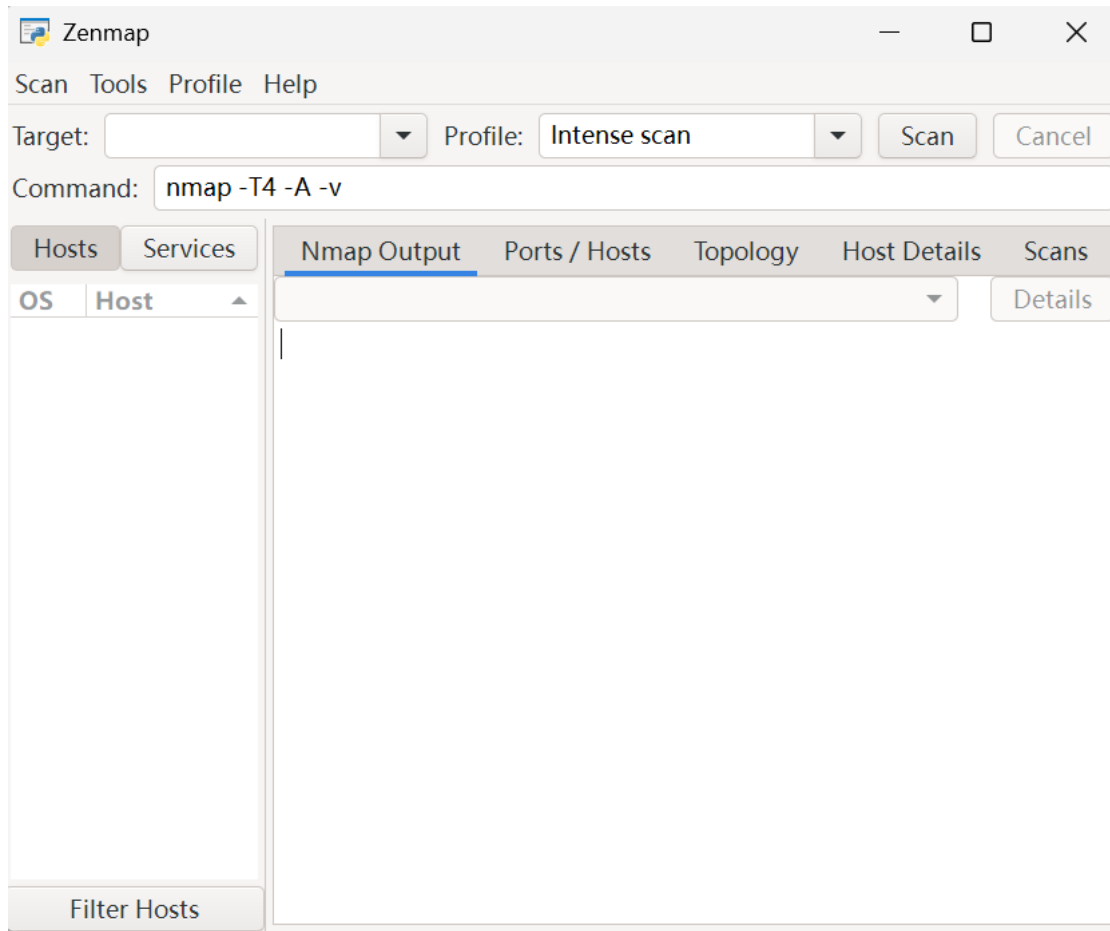
NMap，也就是 Network Mapper，最早是 Linux 下的网络扫描和嗅探工具包。Nmap 是一种目前最强大的信息收集工具，其中综合了各种扫描模式和 OS Fingerprint 技术，以及 TCP 序列号预测难度评估，这种扫描工具不会产生大量的日志记录。它是一款开源的扫描工具，用于系统管理员查看一个大型的网络有哪些主机及其上运行何种服务。它支持多种协议的扫描如 UDP、TCP connect()、TCP SYN(半连接)、Ftp Proxy(暴力攻击)、Reverse-Ident、ICMP(ping Sweep)、FIN、ACK Sweep、Xmas Tree、SYN Sweep 和 Null 扫描。Nmap 还提供一些实用功能如通过 TCP/IP 来鉴别操作系统类型、秘密扫描、动态延迟和重发、平行扫描、通过并行的 ping 鉴别下属的主机、欺骗扫描、端口过滤探测、直接的 RPC 扫描、分布扫描、灵活的目标选择以及端口的描述。Nmap 的特色在于秘密扫描，操作系统探测，多种扫描模式以及指纹识别技术。

Nmap 输出的是扫描目标的列表，以及每个目标的补充信息，至于是哪些信息则依赖于所使用的选项。

三、实验过程

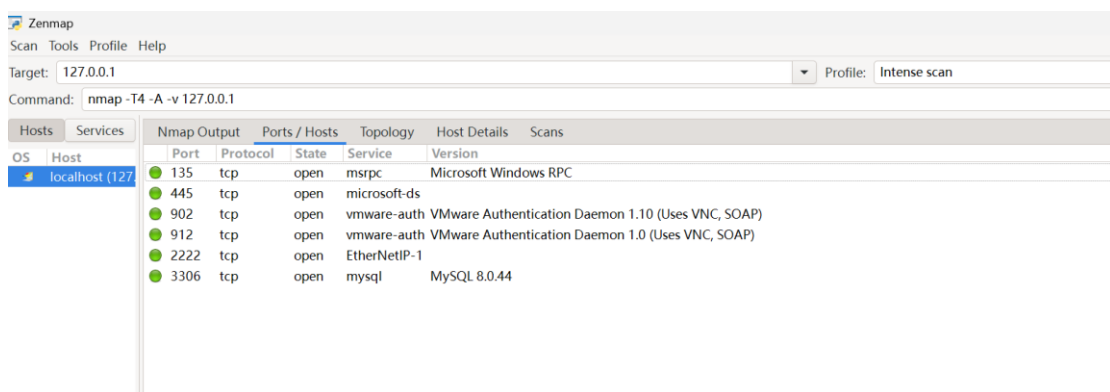
1. 配置 NMap

本次实验我使用 Windows11 完成，首先从官网下载 Nmap 的安装包，按照步骤依次安装 Npcap 和 Nmap 即可。最终打开 GUI 界面如下：



2. Nmap 的扫描实验

首先使用 Nmap 对本地回环地址进行扫描,输入 127.0.0.1 开始扫描。
扫描完成后端口开放情况如下所示:



从图中可以看出,我的电脑开放了 mysql 服务的 3306 端口、虚拟机的端口, 以及 2222 端口用于端口转发。

IP 地址拓扑情况如下所示:

从图中可以看出，起始地址为黑色圆圈，旁边的 localhost 是前面扫描的结果。首先经过 10.136.0.1，这是学校内网的网关地址；接着经过 202.113.20.0/24 网段下的地址和 202.113.18.121，这些都是教育网的公网地址；dns121.online.tj.cn 是天津在线的 DNS 服务器，之后是中国联通的网段；223.120.6.70 和 223.119.66.102 属于中国移动；因为我的域名配置的 DNS 解析是 Cloudflare 的，因此最后的目标归属地属于 Cloudflare，而不会显示域名的源 IP。

下面是输出日志：

```
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-15 13:37 +0800
NSE: Loaded 158 scripts for scanning.
NSE: Script Pre-scanning.
Initiating NSE at 13:37
Completed NSE at 13:37, 0.00s elapsed
Initiating NSE at 13:37
Completed NSE at 13:37, 0.00s elapsed
Initiating NSE at 13:37
Completed NSE at 13:37, 0.00s elapsed
Initiating Parallel DNS resolution of 1 host. at 13:37
Completed Parallel DNS resolution of 1 host. at 13:37, 0.21s elapsed
Initiating Ping Scan at 13:37
Scanning www.djymdd.top (172.67.221.186) [4 ports]
Completed Ping Scan at 13:37, 0.21s elapsed (1 total hosts)
Initiating Parallel DNS resolution of 1 host. at 13:37
Completed Parallel DNS resolution of 1 host. at 13:37, 1.00s elapsed
Initiating SYN Stealth Scan at 13:37
Scanning www.djymdd.top (172.67.221.186) [1000 ports]
Discovered open port 8080/tcp on 172.67.221.186
Discovered open port 80/tcp on 172.67.221.186
Discovered open port 443/tcp on 172.67.221.186
Discovered open port 8443/tcp on 172.67.221.186
Completed SYN Stealth Scan at 13:37, 15.35s elapsed (1000 total ports)
Initiating Service scan at 13:37
Scanning 4 services on www.djymdd.top (172.67.221.186)
Completed Service scan at 13:38, 12.69s elapsed (4 services on 1 host)
Initiating OS detection (try #1) against www.djymdd.top (172.67.221.186)
Retrying OS detection (try #2) against www.djymdd.top (172.67.221.186)
Initiating Traceroute at 13:38
Completed Traceroute at 13:38, 3.06s elapsed
Initiating Parallel DNS resolution of 11 hosts. at 13:38
Completed Parallel DNS resolution of 11 hosts. at 13:38, 1.02s elapsed
NSE: Script scanning 172.67.221.186.
Initiating NSE at 13:38
Completed NSE at 13:39, 64.08s elapsed
Initiating NSE at 13:39
Completed NSE at 13:39, 9.36s elapsed
Initiating NSE at 13:39
Completed NSE at 13:39, 0.01s elapsed
Nmap scan report for www.djymdd.top (172.67.221.186)
Host is up (0.19s latency).
Other addresses for www.djymdd.top (not scanned): 104.21.86.158 2606:4700:3030::ac43:ddba 2606:4700:3031::6815:569e
Not shown: 996 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
80/tcp    open  http    Cloudflare http proxy
|_ http-server-header: cloudflare
|_ http-methods:
|_ Supported Methods: GET
443/tcp   open  ssl/http Cloudflare http proxy
|_ http-server-header: cloudflare
|_ ssl-cert: Subject: commonName=djymdd.top
|_ Subject Alternative Name: DNS:djymdd.top, DNS:*.djymdd.top
|_ Issuer: commonName=WE1/organizationName=Google Trust Services/countryName=US
|_ Public Key type: ec
|_ Public Key bits: 256
|_ Signature Algorithm: ecdsa-with-SHA256
|_ Not valid before: 2026-04-05T10:10:23
|_ Not valid after: 2026-07-04T11:10:16
|_ MD5: 993f 2e61 3a5f 8a25 5867 6d9c 0541 893f
|_ SHA-1: 9f7d 020c f67e 2dbc 0c85 f6f1 0384 dd53 73ee 8764
|_ SHA-256: 39a6 a714 e2c5 e8b5 f5a9 6adb eca3 1d4b 919b aa86 e25b b922 b0c5 b1db 47b3 fe8f
```

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|_http-title: Did not follow redirect to https://dymdd.top/
|_http-methods:
|_Supported-Method: HEAD
|_http-robots.txt: 1 disallowed entry
|_
8080/tcp open  http      Cloudflare http proxy
|_http-server-header: cloudflare
|_http-methods:
|_Supported-Method: HEAD
8443/tcp open  ssl/http Cloudflare http proxy
|_ssl-cert: Subject: commonName=dymdd.top
|_Subject Alternative Name: DNS:dymdd.top, DNS:*.*dymdd.top
|_Issuer: commonName=WSI/organizationName=Google Trust Services/countryName=US
|_Public Key type: ec
|_Public Key bits: 256
|_Signature Algorithm: ecdsa-with-SHA256
|_Not valid before: 2026-04-05T10:10:23
|_Not valid after: 2026-07-04T11:10:16
|_MD5: 993f 2e61 3a5f 8a25 5867 6d9c 0541 893f
|_SHA-1: 5f74 020c 267e 2db0 0c85 f6f1 0384 d653 73ee 8764
|_SHA-256: 3be6 a714 e2c5 e8b5 f5a9 6ed6 eca1 id4b 919e aa8e e25b b922 b0c5 b1db 47b3 fe8f
|_http-server-header: cloudflare
Warning: SSLScan results may be unreliable because we could not find at least 1 open and 1 closed port
Aggressive OS guesses: Apple iOS 14.0 - 15.6 or tvOS 14.3 - 16.1 (Darwin 20.0.0 - 22.1.0) (89%), Apple macOS 11 (Big Sur) - 13 (Ventura) or iOS 16 (Darwin 20.6.0 - 22.4.0) (89%), Apple macOS 14.3 (Sonoma) (Darwin 23.3.0) (89%), Apple macOS 26.1 ( Tahoe) (Darwin 25.1.0) (89%), FreeBSD 11.0-STABLE (89%), Apple iOS 15.7 (Darwin 21.7.0) (88%), Apple iOS 17 (88%), Apple iOS 26.1 (88%), Apple macOS 14.6 (Sonoma) (Darwin 23.6.0) (88%), FreeBSD 11.1-STABLE (88%)
No exact OS matches for host (test conditions non-ideal).
Uptime guess: 0.001 days (since Wed Apr 15 13:38:04 2026)
Network Distance: 14 hops
TCP Sequence Prediction: Difficulty=261 (Good luck!)
IP ID Sequence Generation: All zeros

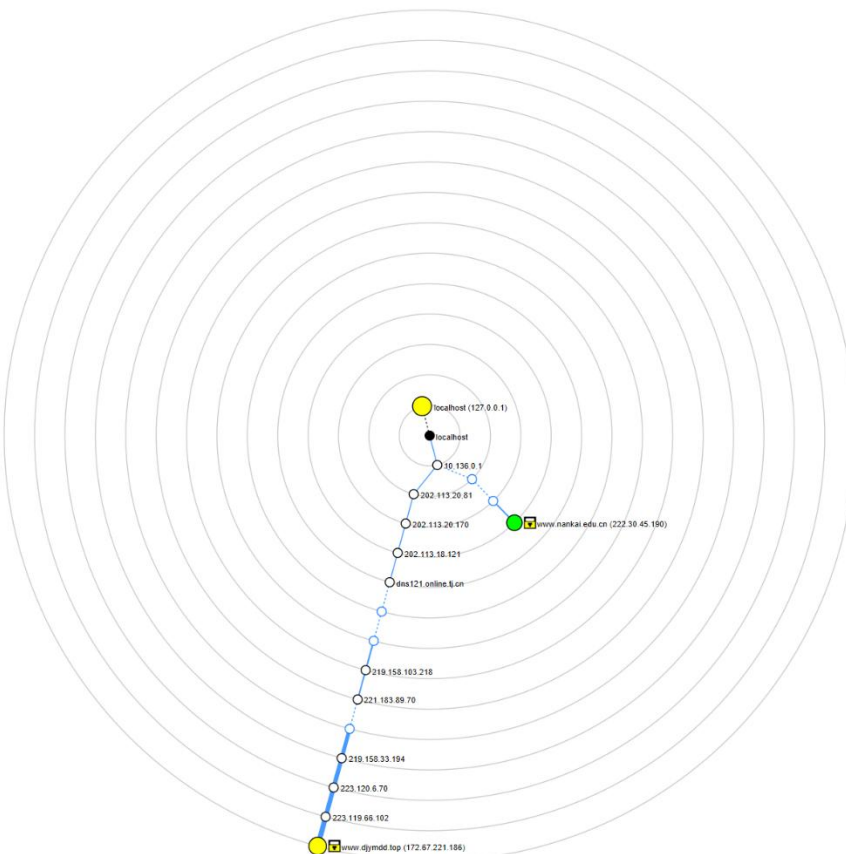
TRACEROUTE (using port 8080/tcp)
  hop  rtg          address
  1    3.00 ms    10.136.0.1
  2    3.00 ms    202.113.20.81
  3    6.00 ms    202.113.20.170
  4    7.00 ms    202.113.18.121
  5    9.00 ms    dns121.online.tj.cn (117.8.157.121)
  6    ... ?
  8    40.00 ms   219.158.103.218
  9    29.00 ms   221.183.89.70
  10   ...
  11   178.00 ms  219.158.33.194
  12   205.00 ms 223.120.6.70
  13   215.00 ms 223.119.66.102
  14   249.00 ms 172.67.221.186

NSE: Script Post-scanning.
Initiating NSE at 13:39
Completed NSE at 13:39, 0.00s elapsed
Initiating NSE at 13:39
Completed NSE at 13:39, 0.00s elapsed
Initiating NSE at 13:39
Completed NSE at 13:39, 0.00s elapsed
Read data files from: D:\Nmap
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/.
Nmap done: 1 IP address (1 host up) scanned in 114.08 seconds
Raw packets sent: 2101 (95.920KB) | Rcvd: 94 (5.220KB)

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从日志中可以看到总共经过了 14 跳，开放的端口有 80、443、8080 和 8443，均是 Cloudflare 的代理端口，证书由 Google Trust Service 签发。

下面对学校官网 www.nankai.edu.cn 进行测试，IP 地址拓扑情况如下所示：



下面是输出日志:

```
443/tcp open  ssl/http-proxy (bad gateway)
```